MUB(6): verified reductions and remaining obligations

Single-agent research consolidation

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AuthorityEffect: None. Author-review draft; no global MUB bound is claimed.

Abstract

This draft integrates the corrected Subproblem 2 body into the inherited seven-section skeleton. Concrete quartet transport, gauge residual identities, and the exact complex Tao calculation have scoped Lean replay receipts. The other sections remain explicit integration or proof obligations. A source-pinned single-Hadamard classification does not supply nonextension of its retained reconstructions. Neither a dimension-wide exclusion nor an exact quartet is established here.

1 Constructive Lower Bound and Projector Gram Spectrum Obstruction

Status. The inherited tensor-product construction supplies the standard three-basis lower-bound route. Its matrices and projector-Gram derivation must be integrated with a fresh, source-bound verification. An obstruction must use information beyond the displayed linear Gram spectrum. *Subproblem 1 body remains pending source-bound integration.*

2 Canonical Gauge, Exact Balances, and Exclusion Obligations

Status. The inserted body separates the twelve-dimensional phase group, 120 ambient coordinates for a fixed first Hadamard, exact balance identities, the Tao Laurent evaluation, and the unresolved exclusion/coverage obligations.

2.1 Normalization and column gauge

Let B_0, B_1, B_2, B_3 be unitary bases of $\mathbb{C}^6$. After common left multiplication by B_0^*, put $H_a = \sqrt{6}\, B_0^* B_a$ for $a = 1, 2, 3$. The quartet equations are

$$|(H_a)_{ij}|^2 = 1, \qquad H_a^* H_a = 6I, \qquad |(H_a^* H_b)_{ij}|^2 = 6 \quad (a \neq b).$$

Fix H_1. Right multiplication of H_2, H_3 by independent diagonal unitaries preserves these equations. Choosing diagonal entries $\overline{(H_a)_{0j}}$ fixes both first rows to one. The acting phase group is $\mathbb{T}^{12}$, of real dimension twelve. The gauge fixes twenty-four ambient real coordinates and also discharges twelve already-imposed circle equations. Thus $144 - 24 = 120$ ambient real coordinates remain. This is not a claim about solution-set dimension, smoothness, or Jacobian rank. If the dephased H_1 varies, its 5×5 block contributes fifty additional ambient coordinates. Concrete normalization and simultaneous monomial transport were replayed in `MubQuartetTransportBridge.lean` and `MubMonomialTransport.lean`. Matching an executable coordinate contractor to this gauge remains a separate obligation.

2.2 Balance equations are ideal consequences

Write $A = H_2$, $B = H_3$, $W = A^*B$, and define

$$g_{A,j} = A_{0j} - 1, \qquad R_{B,m} = \sum_k B_{mk}\overline{B_{0k}} - 6\delta_{m0}.$$

For arbitrary complex matrices, before imposing any equations,

$$\sum_k W_{jk} - 6 = \sum_m \overline{A_{mj}} R_{B,m} - \sum_{m,k} \overline{A_{mj}} B_{mk} \overline{g_{B,k}} + 6\overline{g_{A,j}},$$
$$\sum_j W_{jk} - 6 = \sum_m \overline{R_{A,m}} B_{mk} - \sum_{m,j} g_{A,j} \overline{A_{mj}} B_{mk} + 6 g_{B,k}.$$

The first identity follows by substituting $\overline{B_{0k}} = 1 + \overline{g_{B,k}}$; the second is its conjugate with A, B interchanged. Their real and imaginary parts lie in the ideal generated by the corresponding gauge and row-Gram residuals. Square-matrix column unitarity implies row unitarity, so all twelve complex balance equations vanish on the quartet variety. Adding these consequences does not restrict that variety further, although it may strengthen an interval contractor. No independence claim for the other equations follows. The two identities and ideal-membership corollaries were replayed over an arbitrary commutative star ring in `MubGaugeSyzygy.lean`.

2.3 Exact Tao evaluation and exclusion boundaries

Let $\omega = -1/2 + i\sqrt{3}/2$ and $S_{rj} = \omega^{E_{rj}}$, where

$$E = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 0 & 1 & 2 \\ 0 & 2 & 2 & 1 & 0 & 1 \\ 0 & 2 & 1 & 2 & 1 & 0 \end{pmatrix}.$$

For every sign vector $\gamma \in \{-1, 1\}^6$ with three positive entries,

$$\left| \sum_{j=0}^{5} \prod_{r=0}^{5} S_{rj}^{\gamma_r} \right|^2 = 9.$$

All twenty masks were checked exactly; `MubTaoLaurent.lean` connects the integer computation to complex multiplication and Laurent powers. This evaluation does not establish a vanishing condition on arbitrary MUB triples or independently certify Tao nonextension.

Jaming et al., arXiv:0902.0882v2, exclude a quartet containing the identity and a member of their continuous Fourier family. Brierley and Weigert, arXiv:0901.4051v2, Section 4.1, report the Tao common-vector obstruction. Applying either result requires exact representative matching and the appropriate common-unitary/monomial transport. The inherited unspecified Haagerup ceiling is not admitted without concrete entries and a source theorem. Maxwell and Brierley, arXiv:1402.4070v2, treat the full three-parameter Karlsson family and a Laurent identity; vanishing alone gives no nonextension. The inherited five-family completeness premise is quarantined. A finite-corner classification concerns individual Hadamards and still requires exclusion for every retained reconstruction over its continuous seed domain. It does not by itself bound the number of mutually unbiased bases.

Claim scope. These results support concrete gauge transport, exact ideal consequences, and the displayed Tao evaluation. They do not support $N(6) = 3$, global nonextension, a five-family cover, or a 120-dimensional solution manifold. The narrowest claim is the stated matrix identities and exact evaluation. The next decisive test is a quantified exclusion on the complete reconstruction domain, or an exact quartet witness. This section is one agent's consolidation; the five staged merge assignments are not represented as executed reviews.

3 Continuous-Family Nonextension Obligations

Status. Whole-family nonextension remains open. The two-circulant equations have concrete formal reductions, but regular charts, cusp regions, singular branches, and every candidate basis still need a complete exclusion. The full Karlsson family has three parameters. No five-family partition is assumed. *Subproblem 3 body remains pending source-bound integration.*

4 Polar-Kernel Block Compression onto $U(2)$ and Deficit Contraction Bounds

Status. The proposed polar reduction needs the actual block Gram identities, quantified invertibility and trace bounds, a complete seed cover, and transport back to every candidate. Numerical constants in inherited plans are targets, not validated uniform estimates. *Subproblem 4 body remains pending source-bound integration.*

5 Dual Hyperplane Separation on $U(2)$ with Certified Taylor Remainder Enclosures

Status. Dual separation requires rigorously outward-rounded remainders on each covered cell and a proof that the entire candidate domain is covered. A floating-point dual vector or local margin does not supply this certificate. *Subproblem 5 body remains pending source-bound integration.*

6 Distinct Label Metric Repulsion and High-Resolution Phase Discretization

Status. Metric repulsion must be proved for the actual candidate balls, including their full volumes. Diagonal corners, boundaries, and overlapping labels require explicit treatment. A chosen grid resolution alone does not establish completeness. *Subproblem 6 body remains pending source-bound integration.*

7 Bipartite Graph Clique Exclusion, Outward Interval Replay, and Epistemic Envelope

Status. A graph exclusion is decisive only after sound candidate enclosure, complete coverage, and exact graph/reflection replay. Historical execution records do not become certificates by being counted. A final maximum-three conclusion additionally requires the constructive lower bound. *Subproblem 7 body remains pending source-bound integration.*